Digital Inclusion 2031: Non-Response Risk Index for the ONS Census

Live client project for the Office for National Statistics. Prioritising field resource across all 328 local authorities in England and Wales.

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Team of 6
BEBA live client project · MSc Business Analysis and Consulting

THE QUESTION

Which local authorities should the ONS prioritise for field support in 2031, when the only available dataset is 2021 online response rates?

1 The dataset does not measure what it appears to measure

Paper-first allocation, which is a policy decision rather than resident behaviour, explains 96% of the variation in online response ($R^2 = 0.964$, n = 328). Ranking authorities on the raw figures would have reproduced the ONS's own 2021 targeting decisions and called it analysis.

2 So the index was built on the residual, not the raw rate

A six-component composite risk index was constructed on under-performance against what paper-first allocation already predicts, then stress-tested against 1,000 random weightings so the shortlist survives disagreement about how the components should be weighted.

3 Five authorities, each failing in a different way

Boston, Blaenau Gwent, Kingston upon Hull, Conwy and North Norfolk emerged as the shortlist, deliberately selected to represent distinct and actionable modes of failure rather than five versions of the same problem.

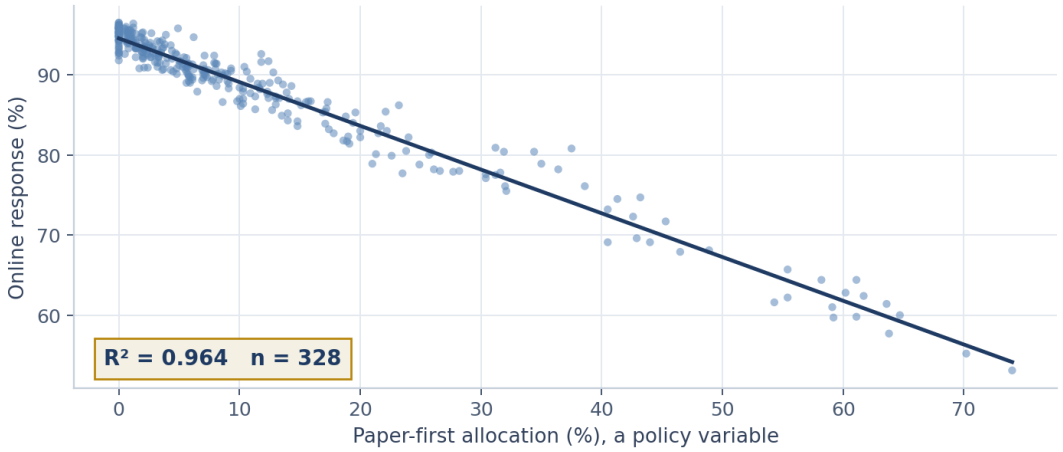
4 A high response rate can hide the risk that matters

The census distributes resources on absolute counts, so every resident estimated rather than counted introduces a decade of potential misallocation. The work followed one such resident, a working-age private renter in Kingston upon Hull, through a persona and journey map.

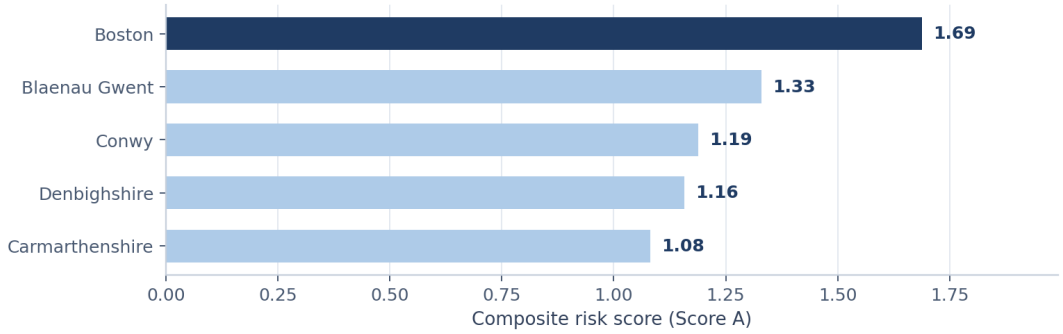
LIMITATIONS, AND WHAT WOULD CHANGE THE ANSWER

The index is built on 2021 data to predict 2031 behaviour, and the relationship between paper-first allocation and online response may not hold under a genuinely digital-first design. The six components were chosen and weighted on judgement; the 1,000-weighting stress test shows the shortlist is not an artefact of one particular weighting, but it cannot show the components are the right ones. The persona is a composite constructed from the data, not an interviewed individual.

The dataset measures policy, not behaviour



Five authorities, five distinct modes of failure



Built with Python (pandas, NumPy, statsmodels) · Excel cross-check · regression · composite indexing

Delivered risk index · five-authority shortlist · persona and journey map · six prioritised recommendations